

PROBLEM SET 5

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: Due on 03/11/2021; scan or type your work and submit through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. Consider the model we saw in lecture 7 about credit rationing. For completeness sake, I write here again the main description. Consider a population of risk-neutral borrowers who each own a project that requires an initial outlay of $I = 1$ and yields a random return X , where $X \in \{0, R\}$. Let $p \in [0, 1]$ denote the probability that $X = R$. Borrowers have no wealth and must obtain investment funds from a bank. A borrower can be of two different types $i = s, r$. The borrower of type i has project with return characteristics (p_i, R_i) . Assume the following.

A1' $p_r R_r < 1 < p_s R_s$.

A2 $p_s > p_r$ and $R_s < R_r$.

As you can see, assumption A1' is different from the one we covered in class. Assume that there is excess demand for funds for the bank, i.e. $\alpha < 1$ and that available funds are sufficient to avoid crowding out either type of borrower completely, i.e. $\max\{\beta, 1 - \beta\} < \alpha$.

- (a) Compute the optimal single fixed repayment scheme D for the bank. Is there any type of borrower that is excluded for sure?
 - (b) Now consider that the bank can offer a type-dependent contract that specifies the probability of financing, x_i and the repayment amount D_i if the project is successful. That is, the bank can offer a plan (x_i, D_i) for $i = s, r$. Compute the optimal contracts for the bank under assumptions A1' and A2. Show that in contrast with what we saw in class, now credit rationing reappears under two possible forms:
 - If $R_s[\beta p_s + (1 - \beta)p_r] \geq 1$ then both types of project are funded and there is cross-subsidization between types. Conclude that this happens because the banks cannot turn away risky borrowers without restricting access to credit for the safe borrowers.
 - If $R_s[\beta p_s + (1 - \beta)p_r] < 1$ then neither project is funded, which is a severe form of credit rationing, i.e. *total financial collapse*.
2. Wuzzle Airlines is the only airline allowed to fly between the Isle of Wuzz and the country of Panda-La. There are two types of passengers, tourists and business. Business travelers are willing to pay more than tourists. The airline, however, cannot tell directly whether a ticket purchaser is a tourist or a business traveler. The two types differ, though, in how much they are willing to pay to avoid having to purchase their tickets in advance. (Passengers do not like to commit themselves in advance to traveling at a particular time.)

More specifically, the utility levels of each of the two types net of the price of the ticket, P , for any given amount of time W prior to the flight that the ticket is purchased are given by

$$\text{Business: } v - \theta_B P - W$$

$$\text{Tourist: } v - \theta_T P - W,$$

where $0 < \theta_B < \theta_T$. (Note that for any given level W , the business traveler is willing to pay more for his ticket. Also, the business traveler is willing to pay more for any given reduction in W .)

The proportion of travelers who are tourists is λ . Assume that the cost of transporting a passenger is c .

Assume in (a)–(d) that Wuzzle Airlines wants to carry both types of passengers.

- (a) Draw the indifference curves of the two types in (P, W) -space. Draw the airline's isoprofit curves. Now formulate the optimal (profit-maximizing) price discrimination problem mathematically that Wuzzle Airlines would want to solve. *Hint: Impose non negativity of prices as a constraint since, if it charged a negative price, it would sell an infinite number of tickets at this price. Waiting times also need to be nonnegative.*
 - (b) Show that in the optimal solution, tourists are indifferent between buying a ticket and not going at all.
 - (c) Show that in the optimal solution, business travelers never buy their ticket prior to the flight and are just indifferent between doing this and buying when tourists buy.
 - (d) Describe fully the optimal price discrimination scheme under the assumption that they sell to both types. How does it depend on the underlying parameters λ , θ_B , θ_T and c ? Provide an economic interpretation.
 - (e) Under what circumstances will Wuzzle Airlines choose to serve only business travelers?
3. A government agency writes a procurement contract with a firm to deliver q units of a good. The firm has constant marginal cost c , so that its profit is $P - cq$, where P denotes the payment for the transaction. The firm's cost is private information and the agency's prior belief about it is uniformly distributed on the interval $[\underline{c}, \bar{c}]$. The agency makes a take-it-or-leave-it offer to the firm (whose default profit is zero). Suppose that $B(q)$ denotes the benefit to the agency of obtaining q units. $B(q)$ is a strictly increasing and a strictly concave function of q .
- (a) Characterize the first-best contract.
 - (b) Characterize the second-best contract.